In Llama, Saying "I Don't Know" Tracks Entity Popularity More Than Knowledge

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Abstract

When a language model answers a factual question with "I don't know", is it reporting that it lacks the answer? We cross entity popularity (Wikipedia page views in PopQA) with knowledge labels taken from the model's own sampled answers. In Llama-3.1-8B-Instruct, abstention tracks popularity more than knowledge: about obscure entities it says "I don't know" in 0.43 of replies to questions it answers correctly at least nine times in ten when made to answer, against 0.00 for popular entities, at similar answer entropy. The gap holds with knowledge relabelled from a second prompt, with unknown facts regraded by a larger model, and counting only replies that name no answer. First-token answer entropy instead follows knowledge in four model families (as the labelling implies) and, in Llama, under the independent labels. Gemma-2-9B rarely abstains under the plain prompt, so there its preference cannot be estimated; under a prompt that permits "I don't know", its abstention also tracks popularity more than knowledge. Steering and logit-lens readouts do not isolate a familiarity mechanism.

1 Introduction

A model that says "I don't know" when it lacks a fact avoids the worst hallucinations [Kalai et al., 2025]. Confidence estimates usually read uncertainty over answers [Kadavath et al., 2022, Kuhn et al., 2023, Stolfo et al., 2024], models use such confidence when a prompt offers abstention [Kumaran et al., 2026], and verbal and answer-level uncertainty diverge [Ji et al., 2025]. What each follows is less clear. Models that converge on shared representations [Huh et al., 2024] might agree on answer uncertainty; whether they also agree on what makes them abstain is open.

People's sense that they can answer a question follows how familiar its terms are, not whether the answer is stored [Reder and Ritter, 1992, Schwartz and Metcalfe, 1992]. In language models, "known entity" features suppress a default "can't answer" response, and Lindsey et al. [2025] propose that a familiar name can trigger them without the fact (a *misfire*). Tests of this idea define an entity as known when the model answers about it correctly [Ferrando et al., 2025], so familiarity and knowledge move together. Representation-level work separates familiarity from factual reliability [Brzezinka, 2026a], and unfamiliar fine-tuning data shape hallucination [Kang et al., 2025], but no study has crossed familiarity and knowledge in abstention behaviour.

We cross them. From PopQA [Mallen et al., 2023] we take questions about obscure and popular entities and label each fact known or unknown from the model's own answers [Gekhman et al., 2024]. If abstention follows familiarity, two off-diagonal cells show it: the

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model should answer about popular entities whose facts it lacks (the misfire) and abstain about obscure entities whose facts it knows (the default not released). Our contribution is this popularity-versus-knowledge dissociation in abstention, its checks, and a comparison across families of what abstention and answer uncertainty follow (Sections 3.1 and 3.2); steering and lens analyses then test, and fail to isolate, a mechanism (Section 3.3).

2 Setup

Questions and labels. We use the 16 relations of PopQA. Within each relation, subjects in the bottom and top deciles of Wikipedia page views, a proxy for familiarity, form the *obscure* and *popular* arms. For each model, a greedy answer and ten samples ($T = 0.7$) under a *forced-answer prompt* (the chat template plus the assistant prefix "The answer is") are graded against gold aliases: a fact is *known* if the greedy answer and at least nine samples are correct and *unknown* if no sample is; the band between is dropped. This leaves 2,303, 2,392, 2,282 and 2,378 questions for Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, Gemma-2-9B-it and Mistral-7B-Instruct-v0.3 [Grattafiori et al., 2024, Qwen et al., 2024, Team et al., 2024, Jiang et al., 2023] (cell counts in Appendix A).

Two readouts. *Answer uncertainty* is the first-token entropy under the forced-answer prompt. Because the knowledge labels are sampled from the same distribution, entropy is expected to follow knowledge, so we also relabel knowledge from an independent prompt. *Abstention* needs the option to decline, so it is measured with a *free prompt*, the chat template alone: for up to 100 questions per popularity × knowledge cell (387 in total) we draw three replies ($T = 0.7$, up to 96 tokens). Two LLM judges (Qwen2.5-32B-Instruct, Gemma-2-27B-it) label each reply DONT_KNOW, CLARIFY (asks which entity is meant) or ANSWER, and a reply abstains when both say DONT_KNOW. On Llama they agree at $\kappa = 0.62$ [0.57, 0.67], above a pre-set bar of 0.60; on the other families they do not (Section 3.2). No human validated the judges.

Effects. The popularity effect is the obscure-minus-popular difference in a readout averaged over the two knowledge levels; the knowledge effect is the unknown-minus-known difference averaged over popularity. Intervals are 95% percentile bootstraps over questions within cells, not clustered by relation. Confirmatory analyses followed criterion notes, analysis plans whose hashes were recorded before each run (Appendix H); the rest are post hoc.

3 Results

3.1 In Llama, abstention tracks popularity; answer uncertainty follows knowledge

About obscure entities, Llama says "I don't know" in 0.43 [0.35, 0.51] of replies to questions it knows, against 0.00 about popular entities (Figure 1a), while forced-answer entropy in these two cells is 0.95 [0.83, 1.08] and 0.86 [0.80, 0.92] nats (Figure 1b). Over the whole design the popularity effect on abstention is +0.39 [+0.33, +0.45] and the knowledge effect +0.09 [+0.03, +0.15]. Their difference, +0.30 [+0.20, +0.40], equals obscure-known minus popular-unknown abstention, so it rests on those two cells. It keeps its sign under five other measurements, from +0.13 to +0.44 (Appendix A); the primary measure was designed after the smallest of these was seen. Answer entropy shows the reverse: it rises by +2.14 nats when Llama lacks the fact and by +0.51 when the entity is obscure.

Labels. Relabelling knowledge with a paraphrased question and prefix (fixed in advance) agrees with the original labels at $\kappa = 0.79$. Of the 87 obscure known facts, 27 stay known and abstain at 0.46; the difference is +0.32 [+0.16, +0.49] (+0.31 [+0.15, +0.48] where both labellings agree), and entropy still follows knowledge (knowledge minus popularity +1.40 [+1.02, +1.73]). Popular entities may have valid answers outside PopQA's aliases, so Llama-3.3-70B regraded forced answers by meaning (fixed in advance). It found 9 of the 100 popular unknown facts known and 31 in between; with its labels the difference is +0.23 [+0.12, +0.35].

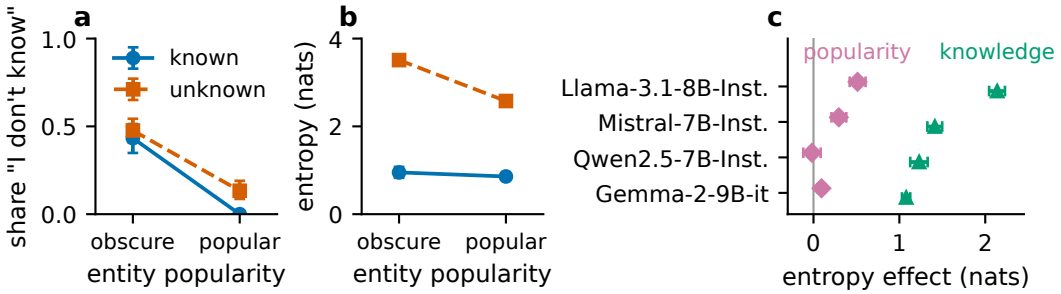


Figure 1: **In Llama, saying "I don't know" tracks popularity, while answer uncertainty follows knowledge.** (a) Llama-3.1-8B-Instruct: share of its free replies (chat template alone, three per question) that both LLM judges label "I don't know" ($n = 387$ questions; obscure known $n = 87$). (b) Its entropy over the first answer token when the reply is forced to begin "The answer is" ($n = 2{,}303$). (c) Popularity and knowledge effects on that entropy in four instruct models. Means with 95% bootstrap intervals over questions.

Measurement. With Llama-3.3-70B as a third judge and a majority rule, $\kappa = 0.74$ and the difference is +0.44 [+0.32, +0.55]. The "I don't know" replies are refusals, not hedged answers: at most 0.03 contain the gold alias in any cell, and counting only replies the 70B judge labels as declining without naming a candidate gives +0.43 [+0.31, +0.54] (fixed in advance).

Alternatives. Llama describes the entity correctly in 0.83 of its completions for obscure known facts and 0.75 for popular known ones; as a covariate, this LLM-graded recognition measure shrinks the popularity effect by 5% (fixed in advance). Ambiguity runs the wrong way (post hoc): popular subjects share their name with more Wikidata entities (median 11–13) than obscure known ones (1), the 73 obscure known facts with a unique name abstain at 0.43, and a namesake-count covariate leaves the popularity coefficient at +0.34 [+0.27, +0.41]. Name form matters more in matched pairs of known facts (gap +0.76 falling to +0.43 [+0.18, +0.69] at equal surface form) than over all questions (Appendix A). Neither model-side measure shows the obscure names to be unfamiliar to Llama: name surprisal is uncorrelated with page views (Appendix F) and recognition is higher for obscure known facts. The result has the shape the misfire idea predicts but does not establish its mechanism.

One prompt. At the forced-answer position, the first token's lean towards hedge tokens follows popularity and knowledge equally (−0.12 [−0.72, +0.45]; Appendix B). After seeing this we read the free prompt: there the log-odds of a first token "I" follow popularity (+8.4 against +3.1) while its entropy follows knowledge (+0.16 against +0.05 nats).

3.2 Shared answer uncertainty; abstention measurable in Llama, and in Gemma when permitted

Entropy follows knowledge in all four families (Figure 1c), as the labelling partly guarantees. Under the free prompt, Qwen, Gemma and Mistral rarely say "I don't know" and the judges disagree on them; with a third judge Gemma reaches $\kappa = 0.64$ but abstains in at most 0.11 of replies in any cell, so its preference cannot be estimated. The pre-registered four-family test on the forced-answer hedge readout returned Qwen PARTIAL, Gemma REPLICATES and Mistral PARTIAL, but that readout shows no preference in Llama either (Appendix B). When the question ends "If you do not know the answer, say that you do not know" (fixed in advance), both models abstain often and three judges agree ($\kappa = 0.71, 0.67$). Popularity minus knowledge is then +0.47 [+0.35, +0.59] for Llama and +0.30 [+0.19, +0.40] for Gemma, which says "I don't know" to 0.41 of obscure known questions and 0.00 of popular known ones. Agreement between models on the forced-answer readout beyond shared labels met a post hoc rule for one of three pairs (Appendix E).

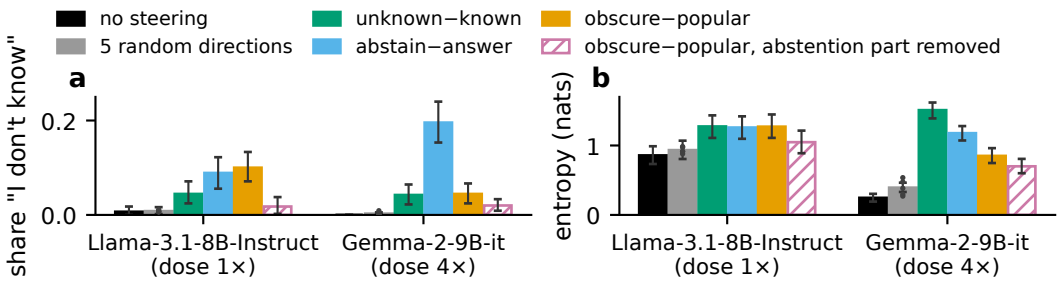


Figure 2: **In Llama, steering along the obscure-minus-popular direction adds little abstention beyond its overlap with an abstention direction.** (a) Share of free replies both judges label "I don't know" on 150 held-out popular questions each model knows; dots: random directions; hatched: obscure-minus-popular with its abstain-minus-answer component projected out. Gemma's two judges agree below the bar; three-judge rates in Appendix C. (b) First-token entropy under the forced-answer prompt. Error bars: 95% bootstrap over questions.

3.3 Mechanistic tests do not isolate a familiarity cause

Steering. As work steering refusal along recognition directions does [Ferrando et al., 2025, Brzezinka, 2026b], we fitted obscure-minus-popular, unknown-minus-known and abstain-minus-answer directions on half of the questions [Arditi et al., 2024, Rimsky et al., 2024], added each at a multiple (the dose) of the obscure-minus-popular direction's norm, chose the dose on a second set and scored a third (Appendix C). The obscure-minus-popular direction raised Llama's abstention from 0.01 to 0.10, +0.09 [+0.06, +0.12] over random directions and short of the pre-set bar of +0.10 (+0.17 with three judges), and Gemma's from 0.00 to 0.04 at 4× the norm (Figure 2a). In Llama it is nearly parallel to the abstention direction (cosine 0.95), partly by construction, since abstaining questions are mostly obscure. With that component removed it adds +0.01 [−0.00, +0.02] (Gemma +0.02 [+0.01, +0.03]). An abstention direction fitted on obscure questions alone (fixed in advance) still has cosine 0.91, and removing it leaves +0.01 [+0.00, +0.03] (three judges). It beats unknown-minus-known on abstention (+0.06 [+0.02, +0.09]), as the cells' rates predict, while in Gemma unknown-minus-known raises entropy +0.66 [+0.58, +0.74] nats more (Figure 2b). Steering thus moves a generic abstention signal [cf. Che and Li, 2026].

Lens. Does a popularity-sensitive hedge signal exist before instruction tuning? With prefitted Jacobian lenses [Gurnee et al., 2026] we read hedge tokens against answer-opening tokens at the forced-answer position in Llama-3.1-8B, Gemma-2-9B, Gemma-2-2B and OLMo-3-7B [Olmo et al., 2025]. In the mid-depth window fixed in advance, popularity raises the hedge-token term in all four, but knowledge raises it at least as much; on the full readout, which the criterion also required, popularity exceeds knowledge in three of four because the answer-token term differs too (Figure 3). Instruct partners lean further towards knowledge on the same readout (Appendix D), so the lens at this position does not show where Llama-Instruct's preference arises.

4 Limitations and conclusion

Abstention is judged by LLMs not validated against human labels. The design compares only the extreme page-view deciles, keeps only facts answered correctly at least nine times in ten or never, and caps cells at 100 questions; models have 2–9B parameters and questions name a single entity. Hedging driven by ambiguity that the judges do not label as clarification is not excluded. In this extreme-contrast design, Llama's answer uncertainty follows knowledge while its abstention tracks popularity more, so in Llama answer uncertainty alone predicts abstention poorly. Across families answer uncertainty agrees, partly by construction; abstention could be compared in two, and when permitted both track popularity more than knowledge.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems 37*, NeurIPS 2024, page 136037–136083. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024. doi: 10.52202/079017-4322. URL http://dx.doi.org/10.52202/079017-4322.

Grzegorz Brzezinka. Does bielik know what it doesn't know? activation dispersion separates entity familiarity from factual reliability across model scale, 2026a. URL https://arxiv.org/abs/2607.07670.

Grzegorz Brzezinka. Graded entity-familiarity readouts in language models: Polish adaptation, cross-language robustness, and refusal steering, 2026b. URL https://arxiv.org/abs/2607.13568.

Tong Che and Yilong Li. Does the model use the feature? separating steering from mechanism in llms, 2026. URL https://arxiv.org/abs/2610.07270.

Javier Ferrando, Oscar Obeso, Senthooran Rajamanoharan, and Neel Nanda. Do i know this entity? knowledge awareness and hallucinations in language models, 2025. URL https://openreview.net/forum?id=WCRQFlji2q.

Zorik Gekhman, Gal Yona, Roee Aharoni, Matan Eyal, Amir Feder, Roi Reichart, and Jonathan Herzig. Does fine-tuning llms on new knowledge encourage hallucinations? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, page 7765–7784. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.444. URL http://dx.doi.org/10.18653/v1/2024.emnlp-main.444.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yeary, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh,

Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti, Vítor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuewei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkang Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippos Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabsa, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser,

Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024. URL https://arxiv.org/abs/2407.21783.

Wes Gurnee, Theo Horsley, Zifan Carl Guo, Tara Rezaei Kheirkhah, Qinyi Sun, Will Hathaway, Neel Nanda, and Dimitris Bertsimas. Universal neurons in gpt2 language models, 2024. URL https://openreview.net/forum?id=ZeI104QZ8I.

Wes Gurnee, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, and Jack Lindsey. Verbalizable representations form a global workspace in language models, 2026. URL https://arxiv.org/abs/2607.15495.

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis, 2024. URL https://openreview.net/forum?id=BH8TYy0r6u.

Ziwei Ji, Lei Yu, Yeskendir Koishekenov, Yejin Bang, Anthony Hartshorn, Alan Schelten, Cheng Zhang, Pascale Fung, and Nicola Cancedda. Calibrating verbal uncertainty as a linear feature to reduce hallucinations. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, page 3769–3793. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.187. URL http://dx.doi.org/10.18653/v1/2025.emnlp-main.187.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mistral 7b, 2023. URL https://arxiv.org/abs/2310.06825.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know, 2022. URL https://arxiv.org/abs/2207.05221.

Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. Why language models hallucinate, 2025. URL https://arxiv.org/abs/2509.04664.

Katie Kang, Eric Wallace, Claire Tomlin, Aviral Kumar, and Sergey Levine. Unfamiliar finetuning examples control how language models hallucinate. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 3600–3612. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.naacl-long.183. URL http://dx.doi.org/10.18653/v1/2025.naacl-long.183.

Asher Koriat and Ravit Levy-Sadot. The combined contributions of the cue-familiarity and accessibility heuristics to feelings of knowing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 27(1):34–53, 2001. ISSN 0278-7393. doi: 10.1037/0278-7393.27.1.34. URL http://dx.doi.org/10.1037/0278-7393.27.1.34.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation, 2023. URL https://arxiv.org/abs/2302.09664.

Dharshan Kumaran, Nathaniel Daw, Simon Osindero, Petar Veličković, and Viorica Patraucean. Causal Evidence that Language Models use Confidence to Drive Behavior, 2026. URL https://arxiv.org/abs/2603.22161.

Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the Biology of a Large Language Model. Transformer Circuits Thread, 2025. URL https://transformer-circuits.pub/2025/attribution-graphs/biology.html.

Yiqi Liu, Yang Wang, Songxin Wang, Chenghao Xiao, and Chenghua Lin. Broken symmetry in llm refusal: Answer release is more local than refusal restoration, 2026. URL https://arxiv.org/abs/2608.15772.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 9802–9822. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.acl-long.546. URL http://dx.doi.org/10.18653/v1/2023.acl-long.546.

Vy Nguyen, Ziqi Xu, Jeffrey Chan, Estrid He, Feng Xia, Renqiang Luo, Erik Cambria, and Xiuzhen Zhang. The commit-abstain circuit: Why language models hallucinate instead of abstaining, 2026. URL https://arxiv.org/abs/2609.32964.

Team Olmo, :, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, Jacob Morrison, Jake Poznanski, Kyle Lo, Luca Soldaini, Matt Jordan, Mayee Chen, Michael Noukhovitch, Nathan Lambert, Pete Walsh, Pradeep Dasigi, Robert Berry, Saumya Malik, Saurabh Shah, Scott Geng, Shane Arora, Shashank Gupta, Taira Anderson, Teng Xiao, Tyler Murray, Tyler Romero, Victoria Graf, Akari Asai, Akshita Bhagia, Alexander Wettig, Alisa Liu, Aman Rangapur, Chloe Anastasiades, Costa Huang, Dustin Schwenk, Harsh Trivedi, Ian Magnusson, Jaron Lochner, Jiacheng Liu, Lester James V. Miranda, Maarten Sap, Malia Morgan, Michael Schmitz, Michal Guerquin, Michael Wilson, Regan Huff, Ronan Le Bras, Rui Xin, Rulin Shao, Sam Skjonsberg, Shannon Zejiang Shen, Shuyue Stella Li, Tucker Wilde, Valentina Pyatkin, Will Merrill, Yapei Chang, Yuling Gu, Zhiyuan Zeng, Ashish Sabharwal, Luke Zettlemoyer, Pang Wei Koh, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. Olmo 3, 2025. URL https://arxiv.org/abs/2512.13961.

Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2024. URL https://arxiv.org/abs/2412.15115.

Lynne M. Reder and Frank E. Ritter. What determines initial feeling of knowing? familiarity with question terms, not with the answer. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 18(3):435–451, May 1992. ISSN 0278-7393. doi: 10.1037/0278-7393.18.3.435. URL http://dx.doi.org/10.1037/0278-7393.18.3.435.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 15504–15522. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.828. URL http://dx.doi.org/10.18653/v1/2024.acl-long.828.

Bennett L. Schwartz and Janet Metcalfe. Cue familiarity but not target retrievability enhances feeling-of-knowing judgments. *Journal of Experimental Psychology: Learning, Memory,*

and Cognition, 18(5):1074–1083, 1992. ISSN 0278-7393. doi: 10.1037/0278-7393.18.5.1074. URL http://dx.doi.org/10.1037/0278-7393.18.5.1074.

Alessandro Stolfo, Ben Wu, Wes Gurnee, Yonatan Belinkov, Xingyi Song, Mrinmaya Sachan, and Neel Nanda. Confidence regulation neurons in language models. In *Advances in Neural Information Processing Systems 37*, NeurIPS 2024, page 125019–125049. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024. doi: 10.52202/079017-3970. URL http://dx.doi.org/10.52202/079017-3970.

Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, page 407–416. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.blackboxnlp-1.25. URL http://dx.doi.org/10.18653/v1/2024.blackboxnlp-1.25.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, Pouya Tafti, Abe Friesen, Michelle Casbon, Sabela Ramos, Ravin Kumar, Charline Le Lan, Sammy Jerome, Anton Tsitsulin, Nino Vieillard, Piotr Stanczyk, Sertan Girgin, Nikola Momchev, Matt Hoffman, Shantanu Thakoor, Jean-Bastien Grill, Behnam Neyshabur, Olivier Bachem, Alanna Walton, Aliaksei Severyn, Alicia Parrish, Aliya Ahmad, Allen Hutchison, Alvin Abdagic, Amanda Carl, Amy Shen, Andy Brock, Andy Coenen, Anthony Laforge, Antonia Paterson, Ben Bastian, Bilal Piot, Bo Wu, Brandon Royal, Charlie Chen, Chintu Kumar, Chris Perry, Chris Welty, Christopher A. Choquette-Choo, Danila Sinopalnikov, David Weinberger, Dimple Vijaykumar, Dominika Rogozińska, Dustin Herbison, Elisa Bandy, Emma Wang, Eric Noland, Erica Moreira, Evan Senter, Evgenii Eltyshev, Francesco Visin, Gabriel Rasskin, Gary Wei, Glenn Cameron, Gus Martins, Hadi Hashemi, Hanna Klimczak-Plucińska, Harleen Batra, Harsh Dhand, Ivan Nardini, Jacinda Mein, Jack Zhou, James Svensson, Jeff Stanway, Jetha Chan, Jin Peng Zhou, Joana Carrasqueira, Joana Iljazi, Jocelyn Becker, Joe Fernandez, Joost van Amersfoort, Josh Gordon, Josh Lipschultz, Josh Newlan, Ju-yeong Ji, Kareem Mohamed, Kartikeya Badola, Kat Black, Katie Millican, Keelin McDonell, Kelvin Nguyen, Kiranbir Sodhia, Kish Greene, Lars Lowe Sjoesund, Lauren Usui, Laurent Sifre, Lena Heuermann, Leticia Lago, Lilly McNealus, Livio Baldini Soares, Logan Kilpatrick, Lucas Dixon, Luciano Martins, Machel Reid, Manvinder Singh, Mark Iverson, Martin Görner, Mat Velloso, Mateo Wirth, Matt Davidow, Matt Miller, Matthew Rahtz, Matthew Watson, Meg Risdal, Mehran Kazemi, Michael Moynihan, Ming Zhang, Minsuk Kahng, Minwoo Park, Mofi Rahman, Mohit Khatwani, Natalie Dao, Nenshad Bardoliwalla, Nesh Devanathan, Neta Dumai, Nilay Chauhan, Oscar Wahltinez, Pankil Botarda, Parker Barnes, Paul Barham, Paul Michel, Pengchong Jin, Petko Georgiev, Phil Culliton, Pradeep Kuppala, Ramona Comanescu, Ramona Merhej, Reena Jana, Reza Ardeshir Rokni, Rishabh Agarwal, Ryan Mullins, Samaneh Saadat, Sara Mc Carthy, Sarah Cogan, Sarah Perrin, Sébastien M. R. Arnold, Sebastian Krause, Shengyang Dai, Shruti Garg, Shruti Sheth, Sue Ronstrom, Susan Chan, Timothy Jordan, Ting Yu, Tom Eccles, Tom Hennigan, Tomas Kocisky, Tulsee Doshi, Vihan Jain, Vikas Yadav, Vilobh Meshram, Vishal Dharmadhikari, Warren Barkley, Wei Wei, Wenming Ye, Woohyun Han, Woosuk Kwon, Xiang Xu, Zhe Shen, Zhitao Gong, Zichuan Wei, Victor Cotruta, Phoebe Kirk, Anand Rao, Minh Giang, Ludovic Peran, Tris Warkentin, Eli Collins, Joelle Barral, Zoubin Ghahramani, Raia Hadsell, D. Sculley, Jeanine Banks, Anca Dragan, Slav Petrov, Oriol Vinyals, Jeff Dean, Demis Hassabis, Koray Kavukcuoglu, Clement Farabet, Elena Buchatskaya, Sebastian Borgeaud, Noah Fiedel, Armand Joulin, Kathleen Kenealy, Robert Dadashi, and Alek Andreev. Gemma 2: Improving open language models at a practical size, 2024. URL https://arxiv.org/abs/2408.00118.

Carter Teplica, Yixin Liu, Arman Cohan, and Tim G. J. Rudner. SCIURus: Shared circuits for interpretable uncertainty representations in language models. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 12451–12469, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi:

10.18653/v1/2025.naacl-long.618. URL https://aclanthology.org/2025.naacl-long.618/.

Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small, 2023. URL https://openreview.net/forum?id=XBflsCjpUM.

A Robustness of the Llama abstention result

Table 1: Popularity effect minus knowledge effect on Llama's abstention under each measurement and labelling. Positive values mean abstention follows popularity more than knowledge. "Corrected": regex rates on 24-token replies corrected for the recall and precision of one LLM judge (Qwen2.5-32B-Instruct), measured on a subset labelled by that judge, not by people. n: questions.

Measurement	Replies	n	Popularity − knowledge
Regex detector	20, 24 tokens	2,303	+0.44 [+0.34, +0.53]
One LLM judge, corrected	same 24-token		+0.13 [+0.03, +0.24]
Two judges, both DONT_KNOW (primary)	3, 96 tokens	387	+0.30 [+0.20, +0.40]
Two judges, DONT_KNOW or CLARIFY	same	387	+0.31 [+0.21, +0.41]
Regex detector	same	387	+0.43 [+0.31, +0.55]
Three judges, majority	same	387	+0.44 [+0.32, +0.55]
70B judge, declines naming no candidate	same	387	+0.43 [+0.31, +0.54]
Two judges, knowledge relabelled (paraphrase)	same	253	+0.32 [+0.16, +0.49]
both labellings agree (post hoc)	same		+0.31 [+0.15, +0.48]
Two judges, unknown facts regraded by meaning	same		+0.23 [+0.12, +0.35]
Two judges, popular unknown with alias in a reply dropped (post hoc)	same		+0.19 [+0.07, +0.31]
Two judges, guessable answers removed (post hoc)	same		+0.30 [+0.19, +0.41]
Two judges, entity recognition as covariate*	same	387	+0.34 [+0.27, +0.41]
Two judges, namesake count as covariate* (post hoc)	same		+0.34 [+0.27, +0.41]

*Coefficient of the obscure arm in a linear probability model with relation fixed effects, not a difference of effects.

Design counts. Llama's pool has 87 known, 85 intermediate and 1179 unknown obscure facts, and 676, 285 and 361 popular ones; the two-judge sample is a stratified random sample of 100 questions per cell (seed 0), or all of a smaller cell. Intervals for differences of cell means use 10,000 within-cell bootstraps (seed 0); linear probability models use 2,000 bootstraps over questions.

Measurements. The single corrected judge gives the smallest difference. Its 24-token replies are often cut off before the model commits; why it and the regex disagree on the same replies is not established. The two-judge measurement with 96 tokens was designed for that reason after the +0.13 was seen and made primary before it ran. Regex recall on the 96-token replies, against the two-judge labels, is 0.99. Each judge's agreement with the three-judge majority: Qwen 0.98, Gemma 0.84, Llama-3.3-70B 1.00.

Independent labels (criterion fixed in advance). Knowledge was relabelled with one paraphrased question per relation, the assistant prefix "Answer:" and fresh samples, under the same rule. Of the 253 questions labelled under both prompts, 62 are known under both, 169 unknown under both, 15 known only originally and 7 only under the paraphrase (agreement 0.91, $\kappa = 0.79$); most other changes move popular known facts to the dropped band. Where both labellings agree, cells have 99 / 27 / 70 / 35 questions (obscure unknown, obscure

known, popular unknown, popular known). Under the new labels, the popularity effect on abstention is +0.38 [+0.29, +0.47] and the knowledge effect +0.06 [−0.03, +0.15].

Regraded unknown facts (criterion fixed in advance). Llama-3.3-70B graded each of a fresh greedy and ten sampled forced answers for every unknown fact as naming the gold entity or not. Of the 100 popular unknown facts in the sample, 60 stay unknown, 9 become known and 31 intermediate (dropped); of the 100 obscure unknown facts, 86, 2 and 12. Dropping instead every popular unknown question with a gold alias in any free reply (0.48 of them) gives +0.19 [+0.07, +0.31].

Strict declines (criterion fixed in advance). The 70B judge labelled each reply as declining without naming a candidate, hedging with a candidate, answering, or asking which entity is meant. Declines occur in 0.66 of obscure known replies and 0.00 of popular known ones; among replies both judges call DONT_KNOW, at most 0.03 per cell are hedged answers.

Entity recognition (criterion fixed in advance). Llama completed "{subject} is" after the question "What is {subject}?" (greedy and five samples), and Llama-3.3-70B graded each completion against the subject's English Wikidata description; recognition is the share of completions graded correct. Its correlation with log page views over all graded questions is $\rho = +0.52$. In the linear probability model its coefficient is −0.03 [−0.06, +0.00] per standard deviation, and it shrinks the popularity coefficient by 5% on the two-judge measure and 3% on the regex measure.

Guessable answers (post hoc). An answer counts as guessable if a gold alias of at least three characters appears in the subject's name, or if it is its relation's most common answer and covers at least 20% of the relation's pool. Among unknown facts alone the popularity gap in abstention is +0.34 [+0.26, +0.43]. On the regex measure over the 2,106 non-guessable questions the difference is +0.48 [+0.36, +0.59].

Namesakes (post hoc). For each subject we counted Wikidata entities with the same English label; medians are 1 (obscure known), 5 (obscure unknown), 11 (popular known) and 13 (popular unknown). The covariate is the standardised log count.

Free replies (post hoc). In the obscure known cell, 0.53 of free replies contain a gold alias, and 0.72 of the replies that do not abstain; for 24 of 87 questions, one free reply gives the correct answer and another says "I don't know". A linear probability model of abstention gives popularity +0.29 [+0.23, +0.36], knowledge −0.05 [−0.14, +0.03] and +0.14 per standard deviation of forced-answer entropy.

Name spelling (post hoc). Over the 2,303 labelled questions, the popularity coefficient on the regex measure moves from −1.87 to −1.79 logits per standard deviation after adding the name's word frequency, token count and token rarity. The matched pairs compare only known facts, one popular with one obscure fact of similar surface form, which is why they lose more of the gap.

B Abstention and the forced-answer hedge readout in all four families

Intervals for the popularity-minus-knowledge rows: three judges, Llama [+0.32, +0.55], Gemma [−0.03, +0.06]; hedge log-odds, Llama [−0.72, +0.45], Qwen [−3.41, −1.64], Gemma [+1.79, +2.42], Mistral [−0.55, +1.34]. The two judges that agree best are Qwen2.5-32B and Llama-3.3-70B ($\kappa = 0.96$ on Llama).

Permissive prompt (criterion fixed in advance). The user message was the question followed by "If you do not know the answer, say that you do not know.", on each model's existing two-judge question set, with three replies scored by the three judges and a majority rule. Cell rates (obscure unknown, obscure known, popular unknown, popular known): Llama 0.84, 0.69, 0.22, 0.00; Gemma 0.54, 0.41, 0.11, 0.00. The criterion let Gemma's verdict replace "cannot be estimated" only if its obscure unknown rate reached 0.25.

Table 2: Judge agreement and abstention. Top: judge agreement and regex recall. Middle: abstention in the obscure, unknown cell. Bottom: three-judge abstention contrast, then effects on first-token hedge-versus-answer log-odds at the forced-answer position.

	Llama	Qwen	Gemma	Mistral
Two judges, κ (bar 0.60)	0.62	0.30	0.47	0.39
Three judges, Fleiss κ (bar 0.60)	0.74	0.41	0.64	0.54
Regex recall against two judges	0.99	0.58	0.48	0.73
Both judges DONT_KNOW	0.48	0.05	0.05	0.10
Both judges DONT_KNOW or CLARIFY	0.65	0.32	0.22	0.25
Three judges, majority	0.75	0.10	0.11	0.19
Regex detector	0.85	0.18	0.07	0.21
Three judges: popularity − knowledge	+0.44	–	+0.01	–
Hedge log-odds: popularity	+2.93	+1.14	+1.79	+2.48
Hedge log-odds: knowledge	+3.05	+3.63	−0.32	+2.08
Hedge log-odds: popularity − knowledge	−0.12	−2.49	+2.11	+0.40
Four-family verdict (pre-registered)	reference	PARTIAL	REPLICATES	PARTIAL

C Steering details

Criterion fixed in advance (Appendix H). Directions were fitted on a random half of each model's questions at layer round$(0.55 \times (L-1))$, the last position of the free prompt: obscure minus popular and unknown minus known (differences of cell means, each averaged over the other factor), and abstain minus answer (questions whose stored free-generation regex abstention rate is at least 0.5 minus those with rate 0). Their natural norms are 4.8, 3.1 and 7.0 in Llama and 62.4, 66.5 and 105.3 in Gemma. The obscure-minus-popular direction with its abstention component projected out was rescaled to the original norm. Each direction was added at a dose $k \in \{0.5, 1, 2, 4, 8\}$ times the obscure-minus-popular direction's natural norm, at the last prompt position and every generated position. k was chosen on 150 popular known questions as the smallest value at which that direction beat five random directions by 0.10 in regex abstention, a rule separate from the confirmatory bar (1× for Llama, 4× for Gemma, whose obscure-minus-popular regex rate was 0.00 at 1× and 0.02 at 2×). All conditions were scored on 150 further popular known questions, 21 of which are also in the two-judge sample.

Table 3: Share of steered replies both judges label "I don't know" (regex detector in parentheses), 150 evaluation questions, 3 replies each.

	None	Random	Unk.−known	Abstain−answer	Obscure−popular	... abstention removed
Llama	0.01 (0.03)	0.008	0.04 (0.10)	0.09 (0.25)	0.10 (0.23)	0.02 (0.02)
Gemma	0.00 (0.00)	0.003	0.04 (0.03)	0.20 (0.74)	0.04 (0.27)	0.02 (0.07)

Primary scorer (both judges), obscure-minus-popular minus random: Llama +0.09 [+0.06, +0.12], Gemma +0.04 [+0.02, +0.06]; with abstention removed: +0.01 [−0.00, +0.02] and +0.02 [+0.01, +0.03]; obscure-minus-popular minus unknown-minus-known: +0.06 [+0.02, +0.09] and +0.00 [−0.03, +0.03]. Secondary scorer (a majority of three judges, added by amendment before judging): obscure-minus-popular minus random +0.17 [+0.13, +0.22] and +0.11 [+0.08, +0.15]; with abstention removed +0.02 [+0.00, +0.04] and +0.05 [+0.03, +0.07]. For Gemma the secondary scorer is the defensible one, since its two judges agree below the bar. The regex detector counts far more abstentions than the judges for steered replies, so an earlier, regex-scored version of this experiment with the dose chosen on the evaluation questions (Llama 0.02 to 0.29) overstated the effect. The cosine between the obscure-minus-popular and abstention directions is 0.95 in Llama and 0.63 in Gemma. No condition produced degenerate text. *Within-level abstention direction* (criterion fixed in advance). To remove the built-in overlap, the abstain-minus-answer direction was refitted on obscure questions only (587 abstaining against 55 answering in Llama, 36 against 520 in Gemma), with the same splits, layer and dose, and scored by a majority of three judges. Its cosine with obscure-minus-popular is 0.91 in Llama and 0.47 in Gemma. Rates for no

steering, the random mean, obscure-minus-popular, the within-level abstention direction and obscure-minus-popular with that direction removed: Llama 0.02, 0.02, 0.19, 0.22, 0.03; Gemma 0.00, 0.01, 0.11, 0.41, 0.03. Over random directions, obscure-minus-popular adds +0.17 [+0.12, +0.22] and +0.10 [+0.07, +0.14], and with the within-level component removed +0.01 [+0.00, +0.03] and +0.03 [+0.01, +0.04]; Llama's pre-registered verdict is that the effect is carried by abstention. A first run launched with the two models' doses swapped produced degenerate Llama text and was discarded before analysis.

D Lens: window effects, the instruct models and onsets

Criterion fixed in advance. Hedge term: log-sum-exp of the lens log-probabilities of the hedge tokens " not", " I", " unknown", " unclear" and " no"; the full readout subtracts the same quantity for the answer-opening tokens " a", " the", " that", " in", " Yes" and " yes". Window: relative depth $\ell/(L-1) \in [1/3, 2/3]$. Intervals: 2,000 within-cell bootstraps. Each model was labelled known or unknown on its own prompt with the same sampling rule.

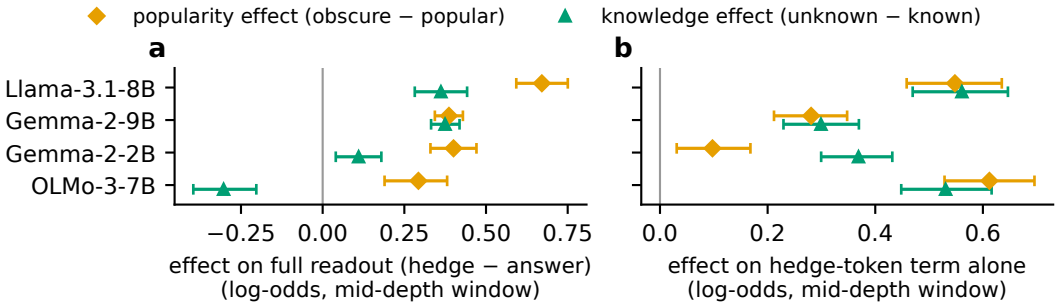


Figure 3: **In base models, the decoded hedge readout responds to popularity, and at least as much to knowledge.** Jacobian-lens readout at the last position of "Question: ... Answer: The answer is", averaged over the pre-set mid-depth window: (a) log-odds of five hedge tokens (e.g. " not", " unknown") against six answer-opening tokens (e.g. " the", " a"); (b) the hedge-token term alone. Points: popularity and knowledge effects with 95% bootstrap intervals. Lenses are fitted per model, so magnitudes are not comparable across models.

Table 4: Mid-depth window effects in base models, with 95% intervals.

	Full readout		Hedge term alone	
Model	Popularity	Knowledge	Popularity	Knowledge
Llama-3.1-8B	+0.67 [+0.59, +0.75]	+0.36 [+0.28, +0.44]	+0.55 [+0.46, +0.64]	+0.56 [+0.47, +0.65]
Gemma-2-9B	+0.39 [+0.34, +0.43]	+0.37 [+0.33, +0.42]	+0.28 [+0.21, +0.35]	+0.30 [+0.23, +0.37]
Gemma-2-2B	+0.40 [+0.33, +0.47]	+0.11 [+0.04, +0.18]	+0.10 [+0.03, +0.17]	+0.37 [+0.30, +0.43]
OLMo-3-7B	+0.29 [+0.19, +0.38]	−0.30 [−0.40, −0.20]	+0.61 [+0.53, +0.70]	+0.53 [+0.45, +0.62]

Instruct partners (criterion fixed in advance). The same readout, window and labelling rule were applied to Llama-3.1-8B-Instruct, Gemma-2-9B-it and Gemma-2-2B-it with their own prefitted lenses and chat-template forced-answer prompts. The criterion allowed the claim that instruction tuning increases the popularity preference only if, for a pair, the instruct difference exceeded the base difference with non-overlapping intervals; this held for no pair.

Onsets. The onset of an effect curve is the first layer where it reaches half its positive maximum over layers and its 95% interval excludes zero. A model supports the prediction that familiarity acts at a shallower depth than knowledge, our layer-depth analogue of a timing account of feelings of knowing [Koriat and Levy-Sadot, 2001], if the knowledge onset follows the popularity onset with an interval above zero, and contradicts it if the interval lies below zero. The prediction was not supported (mixed): 1 supporting, 1 contradicting and 6 undecided of the eight base and instruct models.

Table 5: Popularity minus knowledge in the mid-depth window, base and instruct, with 95% intervals. Lenses differ between the two models of a pair, so compare signs, not magnitudes.

	Full readout		Hedge term alone	
Pair	Base	Instruct	Base	Instruct
Llama-3.1-8B	+0.31 [+0.16, +0.46]	−0.35 [−0.71, +0.01]	−0.01 [−0.17, +0.15]	−1.46 [−1.99, −0.97]
Gemma-2-9B	+0.01 [−0.06, +0.08]	−0.98 [−1.17, −0.78]	−0.02 [−0.15, +0.10]	−1.28 [−1.56, −1.00]
Gemma-2-2B	+0.29 [+0.16, +0.42]	+0.15 [+0.07, +0.22]	−0.27 [−0.38, −0.15]	−0.89 [−1.02, −0.76]

Table 6: Lens onsets (layer index) of the popularity (F) and knowledge (K) effects, their difference $D = K - F$ with its 95% interval, the outcome, and the mid-depth popularity effect on the full readout. Instruct rows use the instruct models' own lenses.

Model	Layers	F onset	K onset	D	Outcome	Mid popularity
Llama-3.1-8B-Instruct	32	17	17	0 [0, 0]	undecided	+1.58
Llama-3.1-8B	32	16	16	0 [0, +1]	undecided	+0.67
Qwen2.5-7B-Instruct	28	20	20	0 [−1, +1]	undecided	+0.21
Gemma-2-9B-it	42	24	24	0 [−2, +1]	undecided	+0.71
Gemma-2-9B	42	24	25	+1 [+1, +2]	supports	+0.39
Gemma-2-2B-it	26	17	15	−2 [−2, 0]	undecided	+0.40
Gemma-2-2B	26	16	16	0 [0, +1]	undecided	+0.40
OLMo-3-7B	32	19	14	−5 [−5, −5]	contradicts	+0.29

E Cross-model agreement (exploratory)

Spearman correlations across questions between Llama's and each other model's forced-answer hedge log-odds and first-token entropies, on questions where both models have the same knowledge label and where they differ (95% bootstrap intervals). The plan, including the rule that hedge agreement and the hedge-minus-entropy gap both have 95% intervals excluding zero, was written before this run, but the point estimates had been seen, so the analysis is post hoc.

	Llama–Qwen	Llama–Gemma	Llama–Mistral
Questions, knowledge differs	313	60	108
Hedge log-odds agreement	+0.11 [−0.00, +0.22]	+0.38 [+0.13, +0.60]	+0.25 [+0.06, +0.42]
Entropy agreement	−0.17 [−0.28, −0.07]	+0.03 [−0.24, +0.28]	−0.07 [−0.27, +0.12]
Gap (hedge − entropy)	+0.28 [+0.14, +0.42]	+0.35 [−0.03, +0.74]	+0.32 [+0.06, +0.57]
Rule	not met	not met	met
Questions, same knowledge	1,784	1,977	2,000
Hedge log-odds agreement	+0.50	+0.38	+0.64
Entropy agreement	+0.46	+0.54	+0.69

F A model-side familiarity measure (negative result)

Criterion fixed in advance. We computed the mean per-token surprisal of each subject name under the base Llama-3.1-8B in the context "Here is a name:". It is uncorrelated with log page views (Spearman −0.001). With knowledge and name length held fixed, its coefficient per standard deviation is +0.013 [−0.026, +0.059] on two-judge abstention and +0.07 [+0.05, +0.10] on the regex measure; adding the popularity arm moves the regex coefficient to +0.01 [−0.01, +0.03], although surprisal hardly correlates with the arm (point-biserial +0.02; −0.06 given relation and knowledge), which we have not resolved. Surprisal therefore does not stand in for popularity. Short names made of common words ("New York" as a book title) are a likely reason.

G Timing, neurons and the circuit

On matched pairs of obscure and popular questions in Llama, the lens-decoded abstention readout reaches half its final size at layer 16, while the entropy gap does so at layer 28, 12 layers later; this compares two readouts of one contrast, not the onsets in Appendix D. Neurons selected for their correlation with output entropy in Llama and Qwen behaved as word-frequency and temperature neurons [Stolfo et al., 2024, Gurnee et al., 2024]. A set of 20 attention heads and 100 MLP neurons found by activation patching [Wang et al., 2023, Syed et al., 2024] reproduces 0.83 [0.78, 0.87] of the first-token abstention log-odds difference when patched from popular into obscure questions and 1.03 [0.99, 1.07] from obscure into popular ones; in generation, removing its state removes abstention in 0.98 of cases but injecting it produces abstention in only 0.40, short of a behavioural bar fixed in advance. Shared uncertainty circuits [Teplica et al., 2025], a commit-or-abstain circuit across ten models [Nguyen et al., 2026] and a similar asymmetry between releasing and restoring refusal [Liu et al., 2026] have been reported before.

H Analysis timing

Each confirmatory analysis had a criterion note written before its run, stating the measurement, the test and the wording each outcome would license; the SHA-256 prefix of each note was recorded before the run, and the notes are in the supplementary material. Four-family design: eba35dec, b53d9a27; one corrected judge: ae0bb807; two-judge measurement: 81acfc9f; lens study: c8e04372; circuit behaviour: 3e1932a9; independent labels: af50f53f; name surprisal: 2b69e06c; lens split: b2b916ce; steering with held-out dose: 18c74883 (amendment adding the third judge as a secondary scorer: 7bed3826; its internal date is a day later than its recorded hash time, a dating error in the note itself); third judge: f35aa4d9; entity recognition: 937b46e3; regraded labels and strict declines: 1498be91; steering with a within-level abstention direction: 05474954; permissive prompt: f6459163; instruct lens: 6af179ce. An earlier steering experiment (da0ad000, 797f52d4) is superseded by the held-out design. The guessability, namesake, spelling, free-reply, entropy-control and label-agreement checks and the cross-model agreement were decided after seeing data and are post hoc.